CPE 314 — Software Design

University of Cebu

Computer Engineering Department

UC-B Campus Marketplace with Reviews

Project Design Document

Table 1: Group information

Group Name / No.	Group 1 / Team 01 - BSCPE 3B - EDP 69323
Members	John Benedict Alaiza
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Revision History

Table 2: Document revisions

Version	Date	Author	Changes
0.1	23 Oct 2026	Group 1	Initial project overview, ER sketch and responsibilities.
0.3	5 Nov 2026	Group 1	API, background JSON import and category charts.
0.3	7 Nov 2026	XionAce	Instructor corrections: one Markdown master, complete ERD attributes, SQL statements, clearer setup and recovery for missing condition details.
0.3	8 Nov 2026	Group 1	Documented the classes, named the IDs and clarified the table connections.

1. Introduction & Project Overview

UC-B Campus Marketplace with Reviews is a local Java desktop app for students, faculty and school groups to post and browse campus goods. Textbook, Electronics and Furniture listings use different condition rules. Members can search categories, join waitlists and review listings. Category summaries show average prices and ratings.

The prototype uses JavaFX, SQLite/JDBC and JSON sales-history files. A member selector lets users browse and sell under a demo profile. A login page and an updated interface are planned.

2. Preliminary ER Sketch & Team Roles

Each box is a database table. A line joins a primary key (PK) to a foreign key (FK). The fork marks the many side of a 1:M relationship. Section 4 shows the complete database.

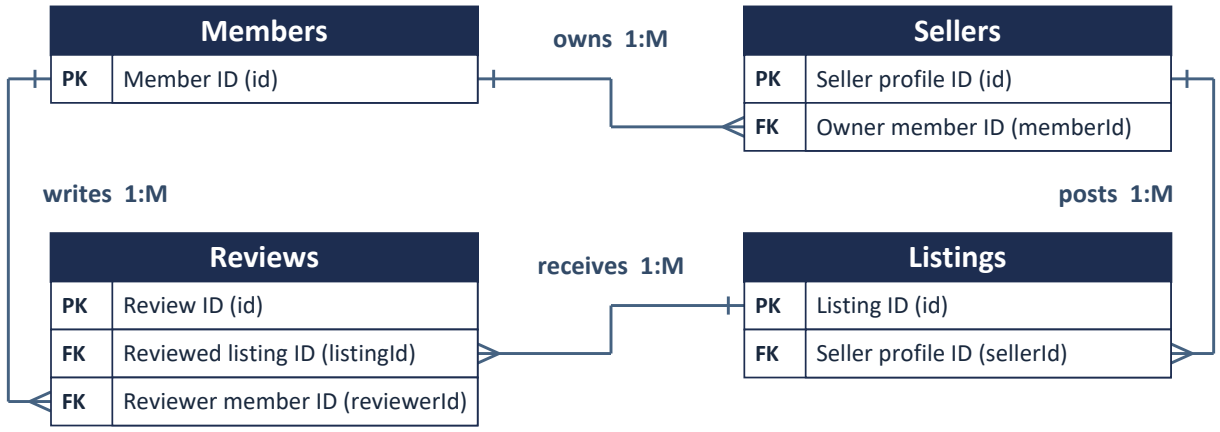


Figure 1: Preliminary ownership and review relationships

Table 3: Team responsibilities

Member and role	Responsibility
John Benedict Alaiza - Backend Developer (Listing Models)	Listing classes, condition rules and validation.
Acexion B. Mahigugmaon - Project Leader / Integration	Connect the components, maintain documents and coordinate fixes.
Glyde V. Arreglado - Database Admin	SQLite tables, saved records and summary queries.
Tristan Dominic Urot - UI/GUI Developer	JavaFX controls, event handlers and charts.
Joshua Brian S. Enovero - Backend & API Developer	Catalog requests, JSON parsing and import.
Emanuelle De Paz - QA & Testing	Test cases, expected calculations and error checks.

3. System Architecture & Object-Oriented Design

Models hold item rules, MarketplaceService coordinates actions, Database handles SQL, and MarketplaceView handles the JavaFX screen. App connects these parts when the program starts.

```
Listing (abstract)
  Textbook extends Listing
  Electronics extends Listing
  Furniture extends Listing
```

The shared Listing fields are ID, seller/category IDs, title, price, currency and status. The abstract methods getType() and gradeCondition() are implemented by each subclass. A List<Listing> can hold all three types. Calling gradeCondition() through Listing runs the actual object's method; this is runtime polymorphism.

Table 4: Model responsibilities and Single Responsibility Principle

Class	Responsibility and reason to change
Listing	Shared listing values and validation. Changes when common listing rules change.
Textbook	Textbook evidence and condition grade. Changes when textbook condition rules change.
Electronics	Electronics evidence and condition grade. Changes when electronics condition rules change.
Furniture	Furniture evidence and condition grade. Changes when furniture condition rules change.
ConditionGrade / WearLevel	Allowed grade and wear values. Changes when these allowed values change.
Member / Category	Member identity and category parent data. Changes when these records change.

Keeping item rules separate from SQL and controls follows the Single Responsibility Principle. A changed grading rule belongs in its item class.

Condition rules

• Textbook: missing pages = POOR; heavy markings = FAIR; light markings or cover damage = GOOD; otherwise LIKE_NEW.

• Electronics: no power or a functional defect = POOR; cosmetic damage = GOOD; otherwise LIKE_NEW.

• Furniture: unstable or structurally damaged = POOR; heavy wear = FAIR; light wear = GOOD; otherwise LIKE_NEW.

Supporting components

Table 5: Supporting responsibilities

Component	Responsibility and reason to change
App	Starts and connects the app; changes when startup setup changes.
MarketplaceView	Controls and event handlers; changes when the screen workflow changes.
MarketplaceService	Coordinates use cases, category search, waitlists and recent views; changes when app rules change.
Database	Schema setup, saved records and SQL summaries; changes when persistence changes.

Collections and recursion

Waitlists use Queue with ArrayDeque: offer() adds at the tail and poll() reads the first buyer. Stored joiningOrder values rebuild FIFO order. A Map stores each member's Stack of recently viewed IDs. push() adds the latest view; pop() reads a copy, preserving session history. At most 20 unique views are kept per member.

collectCategoryIds() adds a category ID and recursively visits each child. A category without children is the base case. Validation rejects missing parents, duplicate IDs and cycles before traversal. Search includes the selected category and all descendants.

Application screen and supporting classes

App opens MarketplaceView. Its controls call MarketplaceService to post, browse, view items and join waitlists. Database saves and loads the records.

ProductCatalogClient requests catalog data. CatalogProduct holds one product. SalesHistoryImporter reads past sales. ProductCatalogClient.Response carries the request result. SalesHistoryImporter.Result reports import counts and errors. Database.CategorySummary carries category averages and counts. Database.ListingForRepair identifies an owner's listing with missing condition data. The owner selects the observed condition and the database restores its missing detail row in a transaction.

GUI workflow and recovery

Marketplace supports posting, searching, viewing, waitlists, reviews and API suggestions. Sales history imports JSON. Analytics shows price and rating charts.

The current busy flag rejects another task while one runs, and action buttons are disabled until it finishes. Members and categories load before the listing search, so a damaged listing cannot block access to recovery controls.

If a listing has no condition-detail row, search reports its ID. Its owner can choose Restore missing details, select the item and explicitly select its observed condition. The saved listing, reviews and waitlist remain. Existing or conflicting details cannot be overwritten through this action; conflicting rows require database-maintainer review. Demo-member selection is not real authentication.

Listing types and sample records

Listing defines the shared item data. The application accepts three concrete types: Textbook, Electronics and Furniture. Database seeds one example of each when it creates the demo data. The user can post additional items of any supported type.

Table 6: Listings supplied with the prototype

Listing and type	Category and price	Saved observations and result
Java textbook (demo) / TEXTBOOK	Textbooks / PHP 350.00	No missing pages, light markings, intact cover. Grade: GOOD.
Gaming laptop (demo) / ELECTRONICS	Gaming Laptops / PHP 18,000.00	Powers on, no functional defect, cosmetic damage. Grade: GOOD.
Study chair (demo) / FURNITURE	Furniture / PHP 650.00	Stable, no structural damage, light surface wear. Grade: GOOD.

Categories and listing rules

All finds is the root category. Its direct children are Textbooks, Electronics and Furniture. Under Electronics, Computers contains Laptops, which contains Gaming Laptops. Searching a parent category includes items in its descendants.

Each new listing requires a seller profile, a category, a nonblank title, a positive price representable in whole minor units and a three-letter uppercase currency. The listing model trims spaces from the title. A post starts as AVAILABLE and uses the active member's seller profile. The service supports owner withdrawal, which changes the status to WITHDRAWN. Marketplace search returns AVAILABLE items.

4. Data Model & Database Design

The schema defines ten application tables. The ERD is split into three views for readability; Listings is repeated to connect the views. Every table attribute is included. TEXT and INTEGER are SQLite types. The schema below specifies nullability and checks.

Read 1:M as one parent to many children. A parent can have no children. M:N (also called M:M) means many-to-many. Members and Listings have this relationship through BuyerInterests for waitlists and through Reviews for ratings. Each connecting table has two 1:M links. Each individual listing subtype uses 1:1; only the matching subtype has a row.

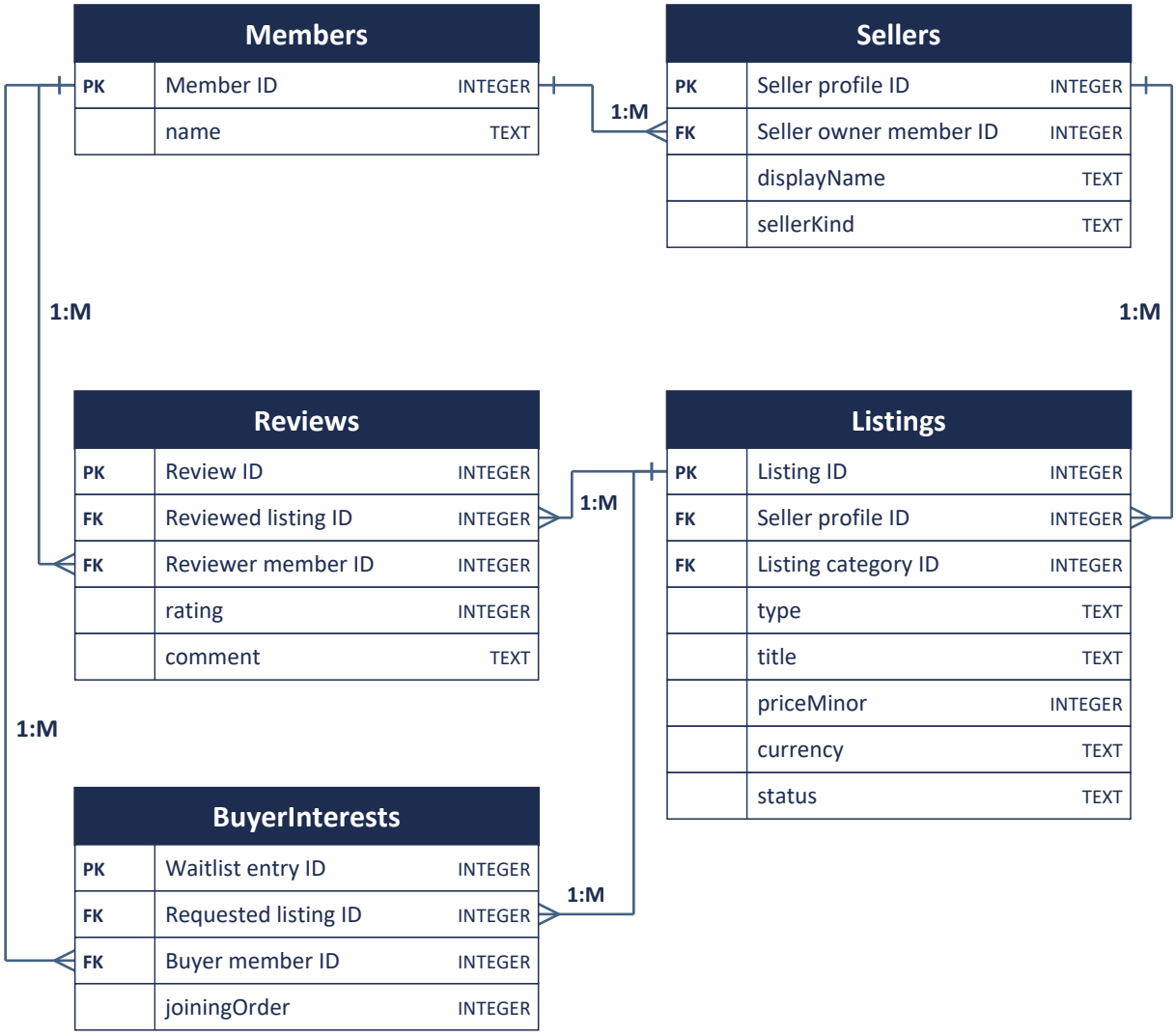


Figure 2: Members, sellers, listings, reviews and buyer interests

Categories and historical sales

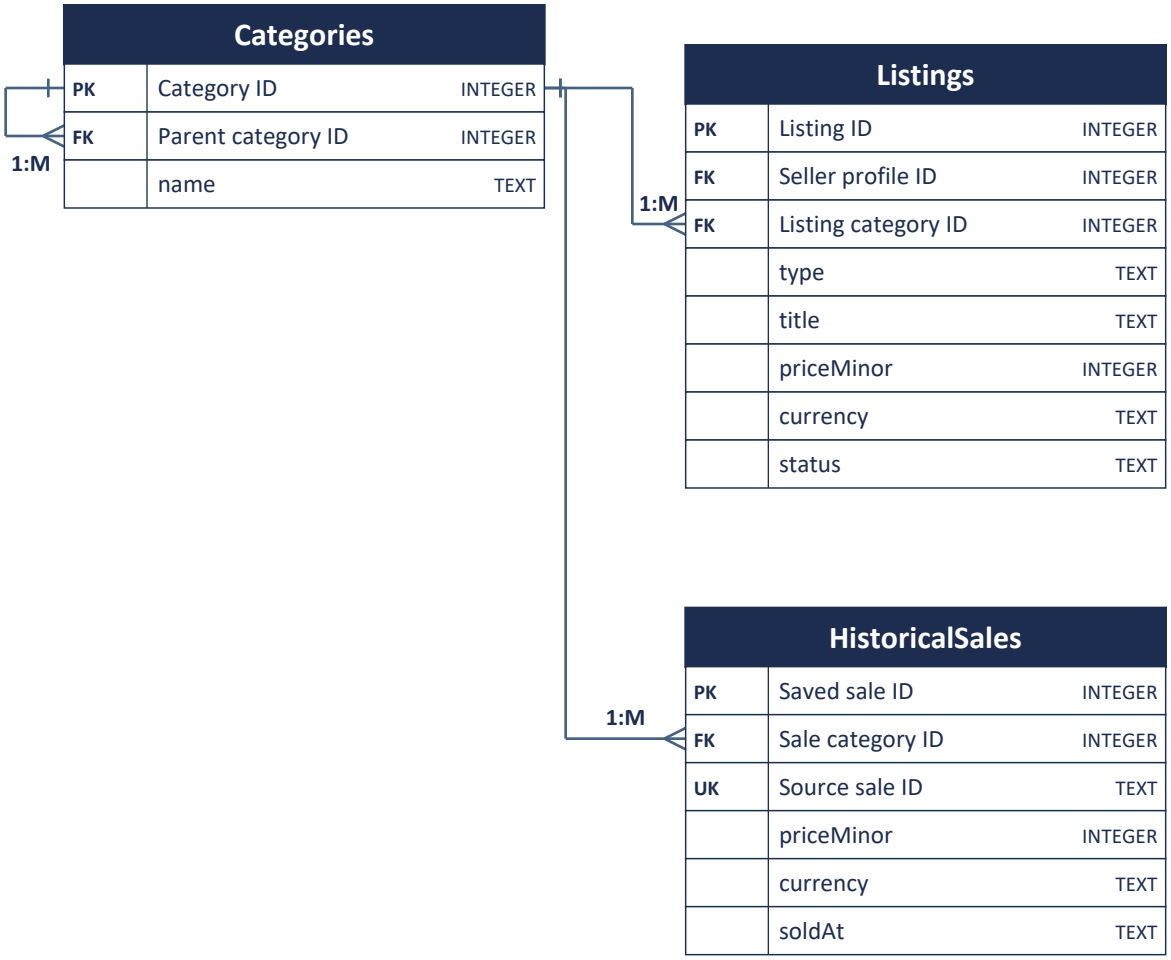


Figure 3: Categories, current listings and imported historical sales

A root category has NULL parentId. A non-root category has one parent, and a parent can have zero or more children. HistoricalSales contains imported past sales, not new marketplace listings. sourceRecordId prevents duplicate imports.

Listing condition details

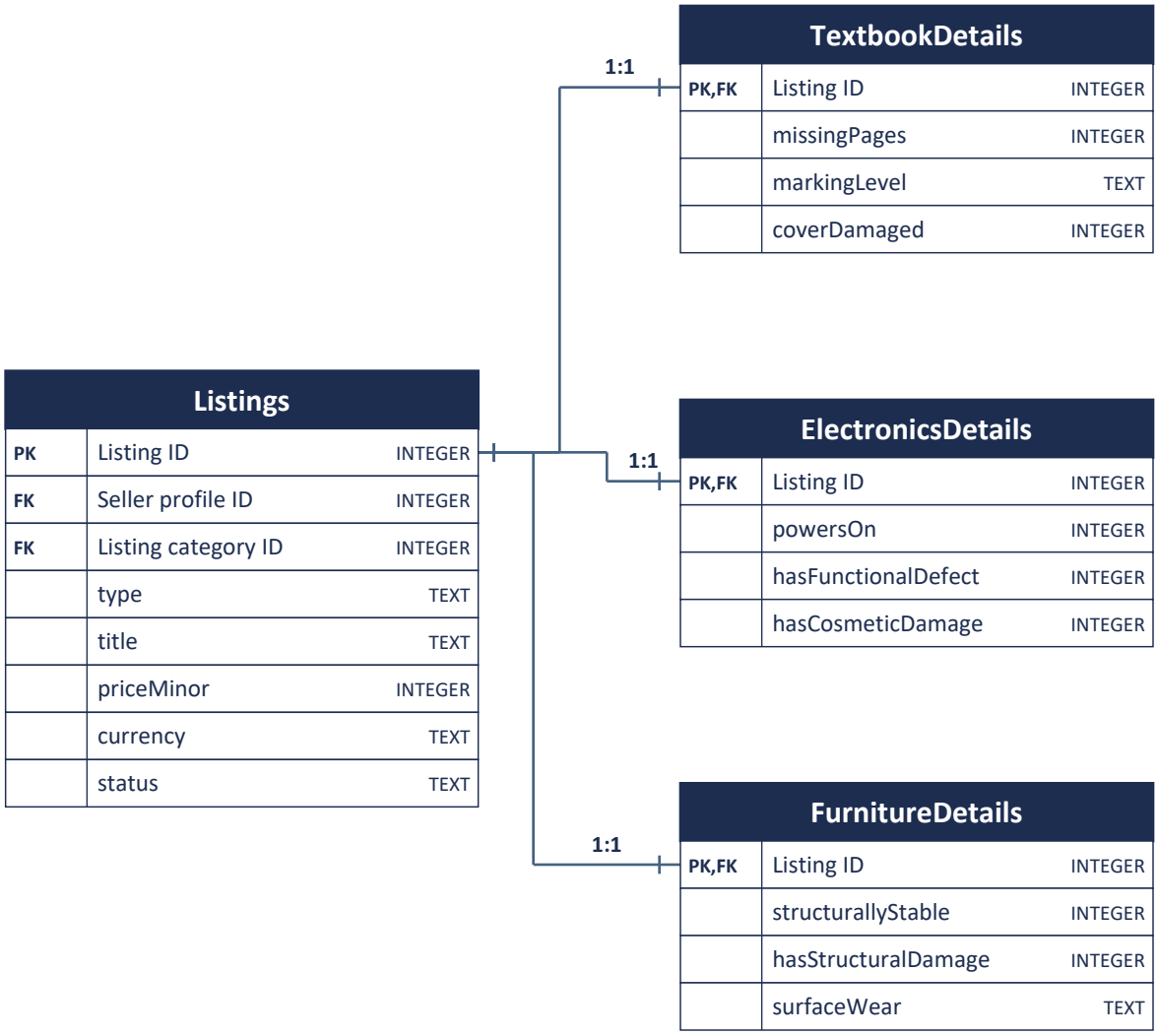


Figure 4: Shared listing fields and type-specific condition evidence

A valid listing needs exactly one matching detail row overall. It has zero or one row in each individual detail table, depending on type. A detail row's listingId is both its primary key and its foreign key. The app saves the listing and detail row in one transaction, and checks the type when reading. Foreign keys alone do not guarantee exactly one matching subtype.

Whose ID is it?

A primary key (PK) identifies one row in its own table. A foreign key (FK) identifies a related row. In the diagrams, Member ID means Members.id, Seller profile ID means Sellers.id, and Listing ID means Listings.id. The SQLite name id always belongs to the table where it appears.

Table 7: IDs used to connect marketplace records

SQLite field	What it identifies
Sellers.memberId	The member who owns the seller profile. Links to Members.id.
Listings.sellerId	The seller profile that posted this item. Links to Sellers.id.
Listings.categoryId	The item's category. Links to Categories.id.
Categories.parentId	The parent category. Links to Categories.id; empty for a root category.
Reviews.listingId	The listing being reviewed. Links to Listings.id.
Reviews.reviewerId	The member who wrote the review. Links to Members.id.
BuyerInterests.listingId	The listing whose waitlist the buyer joined. Links to Listings.id.
BuyerInterests.buyerId	The member who joined the waitlist. Links to Members.id.
HistoricalSales.categoryId	The category of the imported sale. Links to Categories.id.
TextbookDetails.listingId, ElectronicsDetails.listingId, FurnitureDetails.listingId	In each table, this is both PK and FK. It identifies the listing whose condition is described and links to Listings.id.

Example: Review ID 7, Reviewed listing ID 12 and Reviewer member ID 3 mean that member 3 wrote review 7 about listing 12.

HistoricalSales.id identifies the locally saved sale. HistoricalSales.sourceRecordId is the unique ID supplied by the JSON file to recognize a repeat import.

Relationships and storage rules

Table 8: Relationship cardinalities

Parent	Child	Relationship
Members	Sellers	1:M
Members	Reviews	1:M
Members	BuyerInterests	1:M
Sellers	Listings	1:M
Categories	Listings	1:M
Categories	HistoricalSales	1:M
Categories	Categories	1:M; a root has no parent
Listings	Reviews	1:M
Listings	BuyerInterests	1:M
Listings	Each detail table	1:1; only the matching subtype applies

Prices use integer minor units: PHP 350.50 is stored as priceMinor 35050 and currency PHP. Java uses BigDecimal. The app uses AVAILABLE and WITHDRAWN status values. Required names are nonblank; review ratings range from 1 to 5. The application prevents self-review and repeat reviews through the app's serialized transaction path.

Foreign keys are enabled on every SQLite connection. PreparedStatement binds input values to SQL. Each operation opens its own connection. Saving a listing with its detail record and restoring missing details both use transactions and roll back on failure.

The following CREATE TABLE statements match schema.sql. Each snippet defines one table. Index statements remain in the linked schema file.

Members

```
CREATE TABLE Members (
    id INTEGER PRIMARY KEY,
    name TEXT NOT NULL CHECK (length(trim(name)) > 0)
);
```

Sellers

```
CREATE TABLE Sellers (
    id INTEGER PRIMARY KEY,
    memberId INTEGER NOT NULL REFERENCES Members(id),
    displayName TEXT NOT NULL CHECK (length(trim(displayName)) > 0),
    sellerKind TEXT NOT NULL CHECK (length(trim(sellerKind)) > 0)
);
```

Categories

```
CREATE TABLE Categories (
    id INTEGER PRIMARY KEY,
    parentId INTEGER REFERENCES Categories(id),
    name TEXT NOT NULL CHECK (length(trim(name)) > 0),
    CHECK (parentId IS NULL OR parentId <> id)
);
```

Listings

```
CREATE TABLE Listings (
    id INTEGER PRIMARY KEY,
    sellerId INTEGER NOT NULL REFERENCES Sellers(id),
    categoryId INTEGER NOT NULL REFERENCES Categories(id),
    type TEXT NOT NULL CHECK (type IN ('TEXTBOOK', 'ELECTRONICS', 'FURNITURE')),
    title TEXT NOT NULL CHECK (length(trim(title)) > 0),
    priceMinor INTEGER NOT NULL
        CHECK (typeof(priceMinor) = 'integer' AND priceMinor > 0),
    currency TEXT NOT NULL
        CHECK (length(currency) = 3 AND currency NOT GLOB '*[^A-Z]*'),
    status TEXT NOT NULL CHECK (length(trim(status)) > 0)
);
```

Reviews

```
CREATE TABLE Reviews (
    id INTEGER PRIMARY KEY,
    listingId INTEGER NOT NULL REFERENCES Listings(id),
    reviewerId INTEGER NOT NULL REFERENCES Members(id),
    rating INTEGER NOT NULL
        CHECK (typeof(rating) = 'integer' AND rating BETWEEN 1 AND 5),
    comment TEXT
);
```

BuyerInterests

```
CREATE TABLE BuyerInterests (
    id INTEGER PRIMARY KEY,
    listingId INTEGER NOT NULL REFERENCES Listings(id),
    buyerId INTEGER NOT NULL REFERENCES Members(id),
    joiningOrder INTEGER NOT NULL
        CHECK (typeof(joiningOrder) = 'integer' AND joiningOrder > 0),
    UNIQUE (listingId, buyerId),
    UNIQUE (listingId, joiningOrder)
);
```

HistoricalSales

```
CREATE TABLE HistoricalSales (
    id INTEGER PRIMARY KEY,
    categoryId INTEGER NOT NULL REFERENCES Categories(id),
    sourceRecordId TEXT NOT NULL UNIQUE
        CHECK (length(trim(sourceRecordId)) > 0),
    priceMinor INTEGER NOT NULL
        CHECK (typeof(priceMinor) = 'integer' AND priceMinor > 0),
    currency TEXT NOT NULL
        CHECK (length(currency) = 3 AND currency NOT GLOB '*[^A-Z]*'),
    soldAt TEXT NOT NULL CHECK (length(trim(soldAt)) > 0)
);
```

TextbookDetails

```
CREATE TABLE TextbookDetails (
    listingId INTEGER PRIMARY KEY REFERENCES Listings(id) ON DELETE CASCADE,
    missingPages INTEGER NOT NULL
        CHECK (typeof(missingPages) = 'integer' AND missingPages >= 0),
    markingLevel TEXT NOT NULL CHECK (markingLevel IN ('NONE', 'LIGHT', 'HEAVY')),
    coverDamaged INTEGER NOT NULL
        CHECK (typeof(coverDamaged) = 'integer' AND coverDamaged IN (0, 1))
);
```

ElectronicsDetails

```
CREATE TABLE ElectronicsDetails (
    listingId INTEGER PRIMARY KEY REFERENCES Listings(id) ON DELETE CASCADE,
    powersOn INTEGER NOT NULL
        CHECK (typeof(powersOn) = 'integer' AND powersOn IN (0, 1)),
    hasFunctionalDefect INTEGER NOT NULL
        CHECK (typeof(hasFunctionalDefect) = 'integer' AND hasFunctionalDefect IN
        (0, 1)),
    hasCosmeticDamage INTEGER NOT NULL
        CHECK (typeof(hasCosmeticDamage) = 'integer' AND hasCosmeticDamage IN (0,
        1))
);
```

FurnitureDetails

```
CREATE TABLE FurnitureDetails (
    listingId INTEGER PRIMARY KEY REFERENCES Listings(id) ON DELETE CASCADE,
    structurallyStable INTEGER NOT NULL
        CHECK (typeof(structurallyStable) = 'integer' AND structurallyStable IN (0,
        1)),
    hasStructuralDamage INTEGER NOT NULL
        CHECK (typeof(hasStructuralDamage) = 'integer' AND hasStructuralDamage IN
        (0, 1)),
    surfaceWear TEXT NOT NULL CHECK (surfaceWear IN ('NONE', 'LIGHT', 'HEAVY'))
);
```

5. API Integration Design

The app consumes FakeStoreAPI to suggest an electronics category and show comparable catalog prices. Clicking Suggest category / price makes a fresh GET request to https://fakestoreapi.com/products. No API key is required.

Example response shape (illustrative values):

```
[
  {
    "id": 1, "title": "Example monitor", "price": 120.50,
    "category": "electronics", "description": "Product details",
    "image": "https://example.com/product.png",
    "rating": {"rate": 4.2, "count": 25}
  }
]
```

ProductCatalogClient uses Java HttpClient with a 15-second timeout. It checks HTTP 200, parses the JSON with Gson and validates product values. CatalogProduct holds id, title, category and BigDecimal price. Other response fields are not used by this feature.

MarketplaceView calls ProductCatalogClient.matches(), which compares meaningful title words with electronics product titles, ranks matches and returns up to three. The screen selects the local Electronics category and shows the lowest and highest reference prices. The member may choose a child category and enters the final PHP price manually.

The API does not identify the price currency. The app labels the range as a catalog reference with unspecified currency and never silently converts it to PHP. Textbook/furniture or unmatched queries retain manual entry. A timeout, failed status or invalid JSON displays a retry/manual-entry message. A form changed during the request does not receive a stale suggestion. The screen shows request time and HTTP status.

6. Concurrency Design

HTTP requests, large JSON imports and database summaries run on a background worker. MarketplaceView.runTask() creates a JavaFX Task; one ExecutorService executes it. JavaFX reads form values before starting the task. Success and failure callbacks update controls on the JavaFX thread. Platform.runLater() delivers import progress to that thread.

Background JSON import

SalesHistoryImporter streams a UTF-8 JSON array one row at a time. Each record contains sourceRecordId, categoryPath, price, currency and soldAt. Validation checks required values, an existing unique category path, a positive price with exact minor units, an uppercase currency and a timestamp with timezone.

Valid rows go into HistoricalSales. Duplicate source IDs and invalid field values are skipped and counted. One transaction commits the new rows only after the file is read successfully. Broken JSON, database failure or cancellation rolls back the new rows from that import. Existing saved sales remain.

Responsiveness and cancellation

The busy flag rejects overlapping operations. Only work buttons are disabled; tabs, loaded table selection/scrolling, form fields and About remain usable. The import panel displays progress and provides Cancel import. Cancel sets a flag checked between records, and the importer rolls back when it notices the flag. Final saved/invalid/duplicate counts appear after successful completion.

Each database operation owns its connection. Closing the app requests cancellation and shuts down the executor. A short three-record sample finishes quickly; the supplied large fictional JSON generator supports the responsiveness demonstration.

7. Data Mining & Visualization Design

The app summarizes saved marketplace data by category and displays counts alongside averages. The results help a member compare listing prices and campus review ratings.

Table 9: Category calculations

Result	Calculation and data
Average listing price	AVAILABLE Listings in the selected currency, grouped by category: AVG(priceMinor) / 100.0; COUNT(*) gives listing count.
Average review rating	Reviews joined to AVAILABLE Listings, grouped by category: AVG(rating); COUNT(*) gives review count. Highest average identifies the best-rated category; ties are shown.

Separate queries prevent multiple reviews from counting the same listing price repeatedly. Different currencies remain separate. Imported HistoricalSales and external API prices/ratings are excluded from these two calculations. A category without campus reviews is unrated, not zero stars.

Charts and why they fit

Two JavaFX bar charts compare category averages. Each bar represents one category, so bar height makes differences easy to see.

- Average listing price by category: X-axis Category; Y-axis Price in the selected currency, initially PHP.
- Average review rating by category: X-axis Category; Y-axis Rating, on a 0-5 scale.

Both charts have titles and axis labels. Each uses one series, so a legend is unnecessary. Refresh analytics reads current database summaries. Empty results show a no-data title; the screen reports the best-rated category with its review count.

Example: PHP 100 and PHP 200 average PHP 150. Adding PHP 300 makes the average PHP 200. Review ratings 4 and 5 average 4.5. The app calculates these values from saved rows.

Appendix - References and reproduction

The editable master is docs/design-document.md. docs/design-document.pdf is its exported copy. Edit the Markdown first, then run docs/export_documents.py with Python, ReportLab and Calibri fonts. The preliminary sketch uses the linked draw.io file. Section 4 diagrams use the field definitions in this master.

Use JDK 21 for the Java project and Maven launch. The enforcer range [21,22) intentionally rejects JDK 17 and 25. From program/, run `java -version`, `./mvnw.cmd -version`, then `./mvnw.cmd clean verify`. Both version checks must show Java 21; do not skip the enforcer to claim a normal build. The existing Eclipse Java Application and Antigravity Run profiles remain available.

- Team repository
- Project proposal
- Build and demonstration guide
- FakeStoreAPI documentation
- Project Design Document Template
- Week 10 Project Design Document Checklist